

Felix Gröner

Human-LLM interaction researcher designing, developing, and evaluating LLM applications.

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Skills

Research: experiment design, human participants, data science, grant writing

Programming: Python, TypeScript, JavaScript, React, NextJS, SQL, Git

LLM systems: prompt engineering, LangGraph, agent design, RAG

Experience

Research

ADAPT Lab, Arizona State University

2025-2026

Graduate Research Assistant

- Planned and conducted pilot studies and literature reviews.
- Projects:

LLM simulations as synthetic participants in Human Factors research

- Constructed a theory of threats to the validity of LLM simulations.
- Designed and developed LLM agents to act as humanlike study participants.
- Replicated a broad variety of experiments with LLM simulations to compare results.

The role of expertise in human-AI interaction

- Conducted a literature review on empirical investigations of expertise as a factor.
- Developed a framework of types of expertise and potential interactions with other dependent and independent variables.

Center for Accelerating Operational Efficiency

2023-2025

Graduate Research Assistant

- Wrote grant application and research proposals, authored papers.
- Designed, developed, and deployed human-LLM interaction systems.
- Projects:

Graphical user interfaces for directing expectations (\$132,000 awarded)

- Composed a catalog of tasks that elicit misconceptions about LLM capabilities.
- Conducted a literature review of the cognitive psychology of interface design elements (e.g., confidence display variations, cognitive forcing functions).

Reliance on LLM systems in intelligence analysis reporting tasks (\$210,000 awarded)

- Constructed an experimental setup in which participants (n = 304) solved an intelligence analysis task.
- Analyzed the data from different conditions that varied task difficulty, transparency, and stakes.

Human Factors research group, Ulm University, Germany

2022-2023

Undergraduate Research Assistant

- Designed and conducted research studies, project management.
- Projects:

Explainable artificial intelligence (XAI) effects on adherence to AI recommendations

- Conducted an online experiment in which people (n = 404) saw either a global (linear surrogate) or local (LIME) explanation for an AI system's recommendation.
- XAI type did not have as large of an effect as the participants' prior self-perceived expertise.

Technical Projects

Speedrun data analysis and visualization platform

Consumer-facing web application interfacing with public APIs, serving user needs with intuitive and insightful data visualizations.

- Full-stack application built with FastAPI, SQLite, React, Shadcn UI.
- Provides insights into individual patterns, comparisons to other players, and historical developments.

Custom experimentation platform for human-LLM interaction

Survey platform which aims to minimize the friction of using an LLM to solve tasks while recording broad behavioral metrics.

- NextJS frontend and backend, PostgreSQL database, OpenAI API integration.
- Researchers have precise control over the UI and LLM behavior.

Teaching

The Polytechnic School, Arizona State University

2026

Graduate Teaching Assistant

- Developed documentation for a full-stack (Python, Jinja2, MySQL) development capstone project.

The Polytechnic School, Arizona State University

2023

Graduate Teaching Assistant

- Taught Python introductory classes.
- Recorded and edited tutorial videos for Raspberry Pi.

Other

Miller & Meier Consulting, Germany

2021

3-Month Internship

- Identified the workflows and user needs of political consultants.
- Built an NLP prototype to automate part of their daily tasks.

German Embassy to Luxembourg

2019

3-Month Internship

- Organized and attended events with politicians and industry partners.
- Coordinated with international governmental institutions.

Dataport, Germany

2018

Assistant to the Project Management

- Prepared material for stakeholder briefings.

Publications

Peer-reviewed

Gröner, F., Verma, A., & Bronowitz, J. (2026). Automatically Generating Interactive Learning Experiences with an LLM Driven Agentic Pipeline. *Proceedings of the Learning Engineering Research Network Convening (LERN 2026): From Insights to Implementation, Learning Engineering in Action*. <https://doi.org/10.59668/2551.25387>

Gröner, F., & Chiou, E. K. (2024). Investigating the impact of user interface designs on expectations about large language models' capabilities. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. <https://doi.org/10.1177/10711813241260399>

Kraus, J. M., Merger, J., **Gröner, F.,** & Pätz, J. (2023). 'Sorry' says the robot: The tendency to anthropomorphize and technology affinity affect trust in repair strategies after error. *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (pp. 436-441)*. <https://doi.org/10.1145/3568294.3580122>

Preprints

Gröner, F., & Chiou, E.K. (2025). *Engineering approaches to improve human-LLM interaction: A scoping literature review*. <https://ssrn.com/abstract=6505859>

Gröner, F., & Becker, D.V. (2026). *Subjective and objective metrics to explain perceptions of original and AI-generated abstract artworks*. <https://ssrn.com/abstract=6482420>

Gröner, F., & Chiou, E.K. (in preparation). *Threats to the validity of LLM simulations as synthetic participants.*

Gröner, F., & Chiou, E.K. (under review). *A comparison of human and LLM-simulated participants in a writing style task.*

Gröner, F., Mancenido, M. V., Kim, N., Summers, E., & Chiou, E. K. (under review). *Increasing transparency of LLM systems does not always improve people's verification behavior and performance: Results from an empirical study of AI-assisted intelligence analysis reporting.*

Gröner, F., Mancenido, M. V., & Chiou, E. K. (in preparation). *Considering expertise as a factor in the design of synthetic test environments for human-AI research.*

Education

Arizona State University, USA

Expected Fall 2026

Ph.D. Human Systems Engineering

Status: Ph.D. candidate, GPA 4.0

Committee: Erin K. Chiou (Advisor), D. Vaughn Becker, Nancy J. Cooke

Dissertation topic: Engineering and validating LLM agents as synthetic participants for Human Factors research

Ulm University, Germany

June 2024

M.Sc. Cognitive Systems

Thesis: The role of explainable artificial intelligence in trust and adherence to AI recommendations

Kiel University, Germany

December 2022

M.A. International Politics and International Law

Thesis: Necessity of adapting International Humanitarian Law to cyber warfare

Kiel University, Germany

December 2020

B.A. Computer Science and Political Science

Thesis: NATO's strategy to deter cyber attacks

Kiel University, Germany

December 2019

B.Sc. Business Informatics

Thesis: Client's influence on the choice of project management technique in software development